Computer Vision – Lecture 18

Uncalibrated Reconstruction

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Computer Vision

Course Outline

- Image Processing Basics
- Segmentation & Grouping
- Object Recognition & Categorization
- Deep Learning for Vision
- 3D Reconstruction
  - Local Features
  - Epipolar Geometry and Stereo Basics
  - Camera calibration & Triangulation
  - Uncalibrated Reconstruction & Structure-from-Motion

Topics of This Lecture

- Camera calibration
  - Recap: Camera parameters
  - Calibration algorithm
- Revisiting Epipolar Geometry
  - Triangulation
  - Calibrated case: Essential matrix
  - Uncalibrated case: Fundamental matrix
  - Weak calibration
  - Epipolar Transfer
- Sneak Peek: Structure from Motion (SfM)
  - Motivation
  - Ambiguity

Recap: A General Point

- Equations of the form

$$\mathrm{A}\mathrm{x} = 0$$

- How do we solve them? (always!)
  - Apply SVD

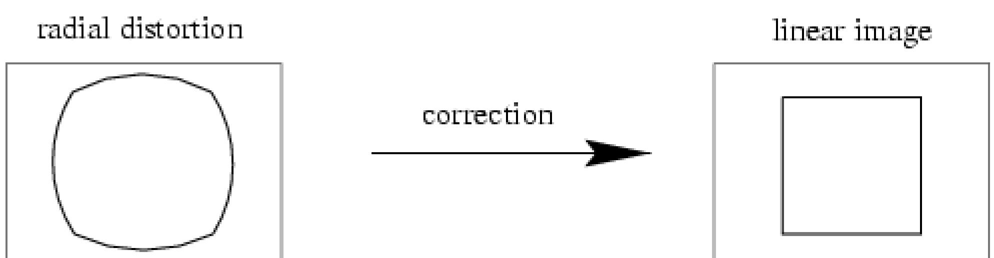
$$\boldsymbol{A} \overset{\text{SVD}}{=} \boldsymbol{UDV}^T = \boldsymbol{U} \underbrace{\begin{bmatrix} d_{11} & & \\ & \ddots & \\ & & d_{NN} \end{bmatrix}}_{\text{Singular values}} \underbrace{\begin{bmatrix} v_{11} & \cdots & \boxed{v_{1N}} \\ \vdots & \ddots & \boxed{\vdots} \\ v_{N1} & \cdots & \boxed{v_{NN}} \end{bmatrix}^T}_{\text{Singular vectors}}$$

  - Singular values of A = square roots of the eigenvalues of A^TA.
  - The solution of Ax=0 is the *nullspace* vector of A.
  - This corresponds to the *smallest singular vector* of A.

Recap: Camera Parameters

- Intrinsic parameters
  - Principal point coordinates
  - Focal length
  - Pixel magnification factors
  - *Skew (non-rectangular pixels)*
  - *Radial distortion*

$$K = \begin{bmatrix} m_x & & \\ & m_y & \\ & & 1 \end{bmatrix} \begin{bmatrix} f & s & p_x \\ & f & p_y \\ & & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ & \alpha_y & y_0 \\ & & 1 \end{bmatrix}$$



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Summary: Camera Parameters

- Intrinsic parameters
  - Principal point coordinates
  - Focal length
  - Pixel magnification factors
  - *Skew (non-rectangular pixels)*
  - *Radial distortion*

$$K = \begin{bmatrix} m_x & & \\ & m_y & \\ & & 1 \end{bmatrix} \begin{bmatrix} f & s & p_x \\ & f & p_y \\ & & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & x_0 \\ & \alpha_y & y_0 \\ & & 1 \end{bmatrix}$$

- Extrinsic parameters
  - Rotation $\mathbf{R}$
  - Translation $\mathbf{t}$
    (both relative to world coordinate system)

- Camera projection matrix

$$\mathrm{P} = \mathrm{K}\left[\mathrm{R} \mid \mathrm{t}\right] = \begin{bmatrix} P_{11} & P_{12} & P_{13} & P_{14} \\ P_{21} & P_{22} & P_{23} & P_{24} \\ P_{31} & P_{32} & P_{33} & P_{34} \end{bmatrix}$$

How many degrees of freedom does P have?

Slide adapted from Svetlana Lazebnik

Camera Parameters: Degrees of Freedom

- Intrinsic parameters — DoF
  - Principal point coordinates — 2
  - Focal length — 1
  - Pixel magnification factors — 1
  - *Skew (non-rectangular pixels)* — *1*
  - *Radial distortion*

$$K = \begin{bmatrix} \alpha_x & s & x_0 \\ & \alpha_y & y_0 \\ & & 1 \end{bmatrix}$$

- Extrinsic parameters
  - Rotation R — 3
  - Translation t — 3

    (both relative to world coordinate system)

- Camera projection matrix

$$P = K[R \mid t]$$

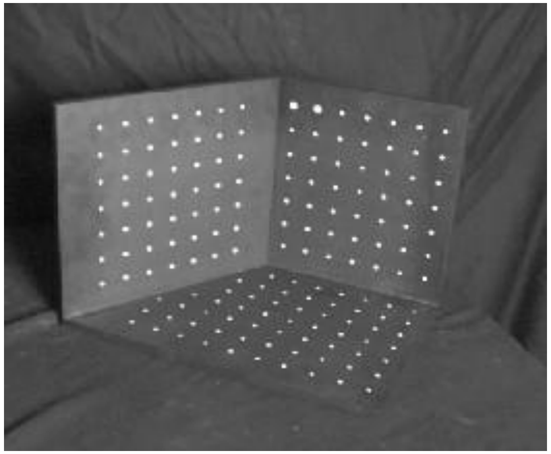
$\Rightarrow$ General pinhole camera: 9 DoF

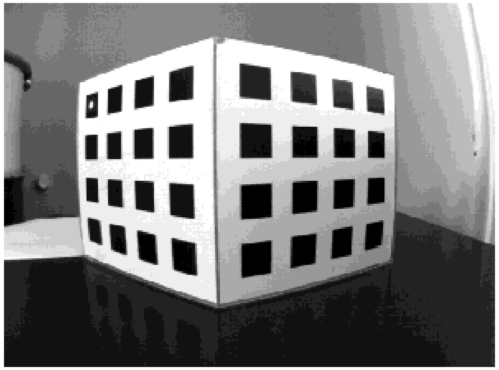
$\Rightarrow$ CCD Camera with square pixels: 10 DoF

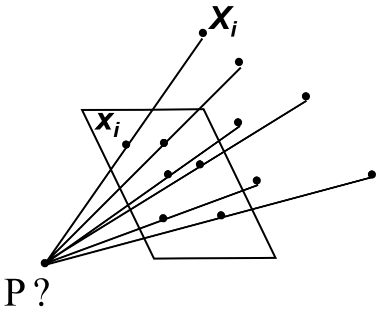
$\Rightarrow$ General camera: 11 DoF

Recap: Calibrating a Camera

- Goal
  - Compute intrinsic and extrinsic parameters using observed camera data.
- Main idea
  - Place "calibration object" with known geometry in the scene
  - Get correspondences
  - Solve for mapping from scene to image: estimate $P = P_{int} P_{ext}$







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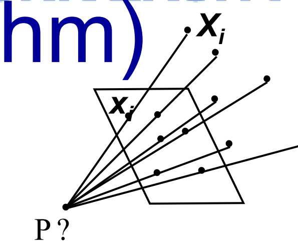
Recap: Camera Calibration (DLT Algorithm)



$$\begin{bmatrix} 0^T & X_1^T & -y_1 X_1^T \\ X_1^T & 0^T & -x_1 X_1^T \\ \cdots & \cdots & \cdots \\ 0^T & X_n^T & -y_n X_n^T \\ X_n^T & 0^T & -x_n X_n^T \end{bmatrix} \begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix} = 0 \qquad \mathbf{A}\mathbf{p} = 0$$

- P has 11 degrees of freedom.
- Two linearly independent equations per independent 2D/3D correspondence.
- (similar to homography estimation)
  - Solution corresponds to smallest singular vector.
- 5 ½ correspondences needed for a minimal solution.

Slide adapted from Svetlana Lazebnik

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Two-View Geometry: 3 Main Questions

- Scene geometry (structure):
  - Given corresponding points in two or more images, where is the pre-image of these points in 3D?
- Correspondence (stereo matching):
  - Given a point in just one image, how does it constrain the position of the corresponding point x' in another image?
- Camera geometry (motion):
  - Given a set of corresponding points in two images, what are the cameras for the two views?

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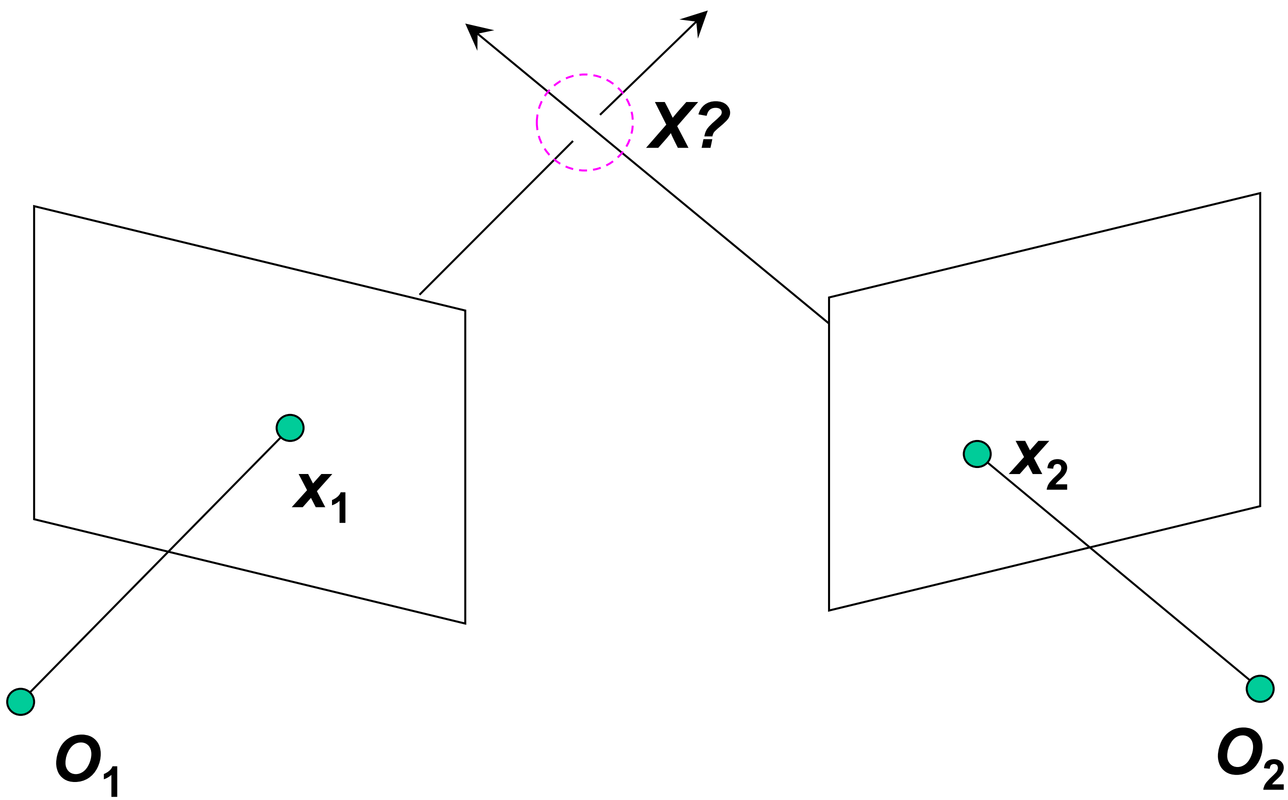
Revisiting Triangulation

- Given projections of a 3D point in two or more images (with known camera matrices), find the coordinates of the point



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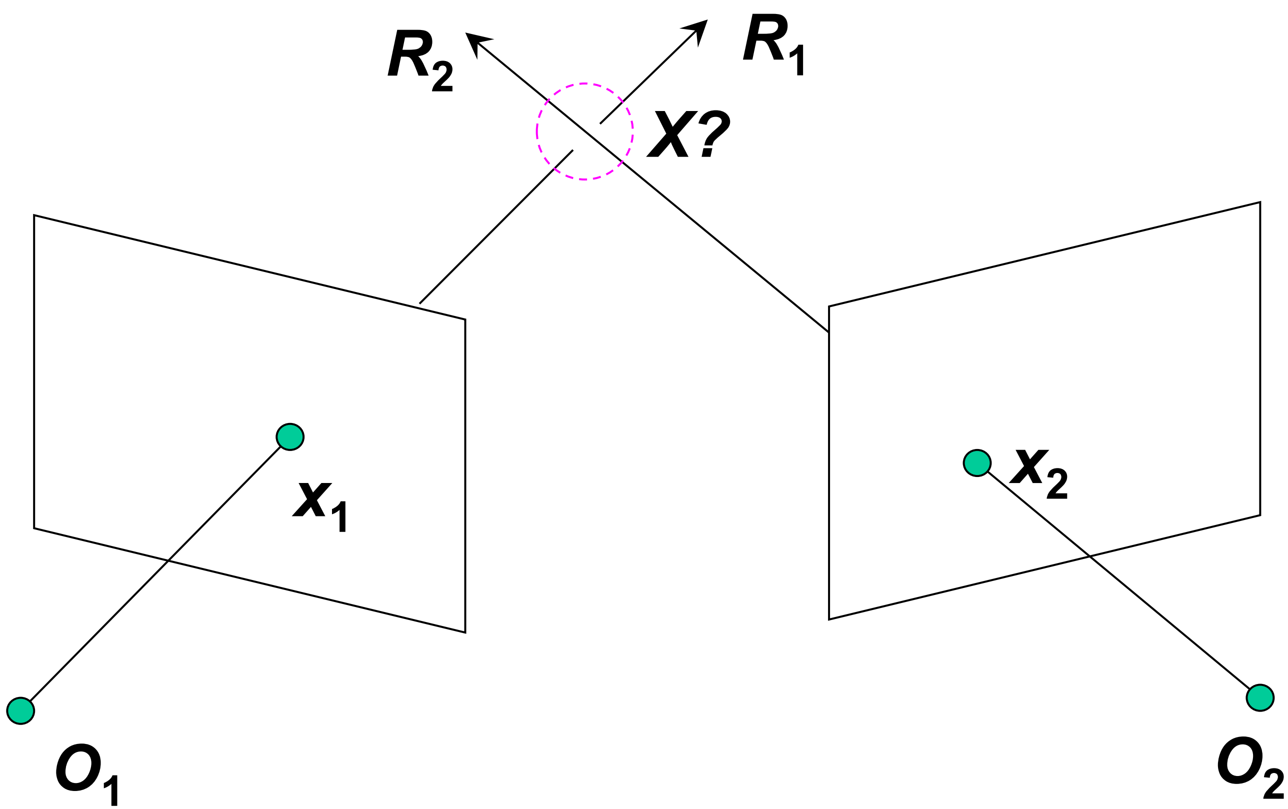
Revisiting Triangulation

- We want to intersect the two visual rays corresponding to x_1 and x_2, but because of noise and numerical errors, they will never meet exactly. How can this be done?



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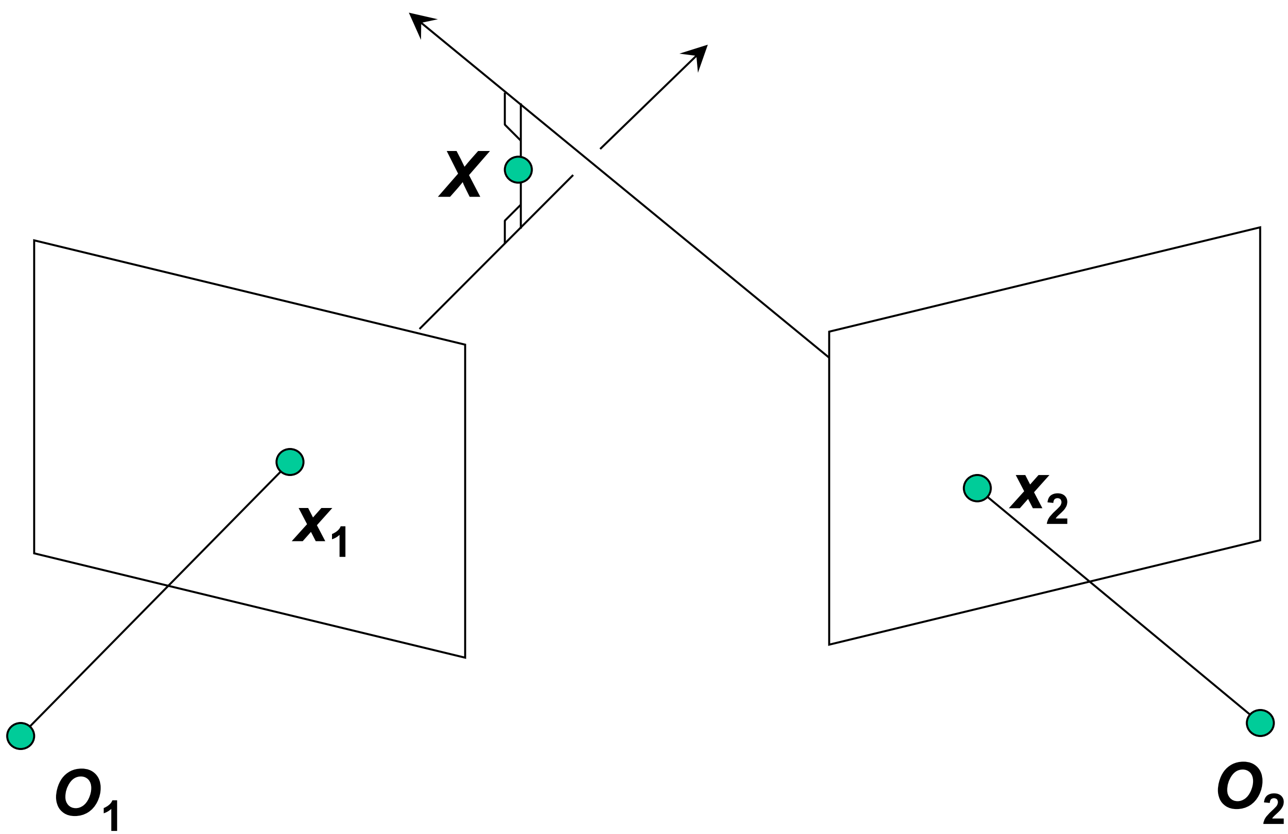
Triangulation: 1) Geometric Approach

- Find shortest segment connecting the two viewing rays and let X be the midpoint of that segment.



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Triangulation: 2) Linear Algebraic Approach

$$\lambda_1 \mathrm{x}_1 = \mathrm{P}_1 \mathrm{X} \qquad \mathrm{x}_1 \times \mathrm{P}_1 \mathrm{X} = 0 \qquad [\mathrm{x}_{1\times}]\mathrm{P}_1 \mathrm{X} = 0$$

$$\lambda_2 \mathrm{x}_2 = \mathrm{P}_2 \mathrm{X} \qquad \mathrm{x}_2 \times \mathrm{P}_2 \mathrm{X} = 0 \qquad [\mathrm{x}_{2\times}]\mathrm{P}_2 \mathrm{X} = 0$$

Cross product as matrix multiplication:

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{bmatrix} \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix} = [\mathbf{a}_\times]\mathbf{b}$$

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Triangulation: 2) Linear Algebraic Approach

$$\lambda_1 \mathrm{x}_1 = \mathrm{P}_1 \mathrm{X} \qquad \mathrm{x}_1 \times \mathrm{P}_1 \mathrm{X} = 0 \qquad [\mathrm{x}_{1\times}]\mathrm{P}_1 \mathrm{X} = 0$$

$$\lambda_2 \mathrm{x}_2 = \mathrm{P}_2 \mathrm{X} \qquad \mathrm{x}_2 \times \mathrm{P}_2 \mathrm{X} = 0 \qquad [\mathrm{x}_{2\times}]\mathrm{P}_2 \mathrm{X} = 0$$

Two independent equations each in terms of three unknown entries of X

$\Rightarrow$ Stack them and solve using SVD!

- This approach is often preferable to the geometric approach, since it nicely generalizes to multiple cameras.

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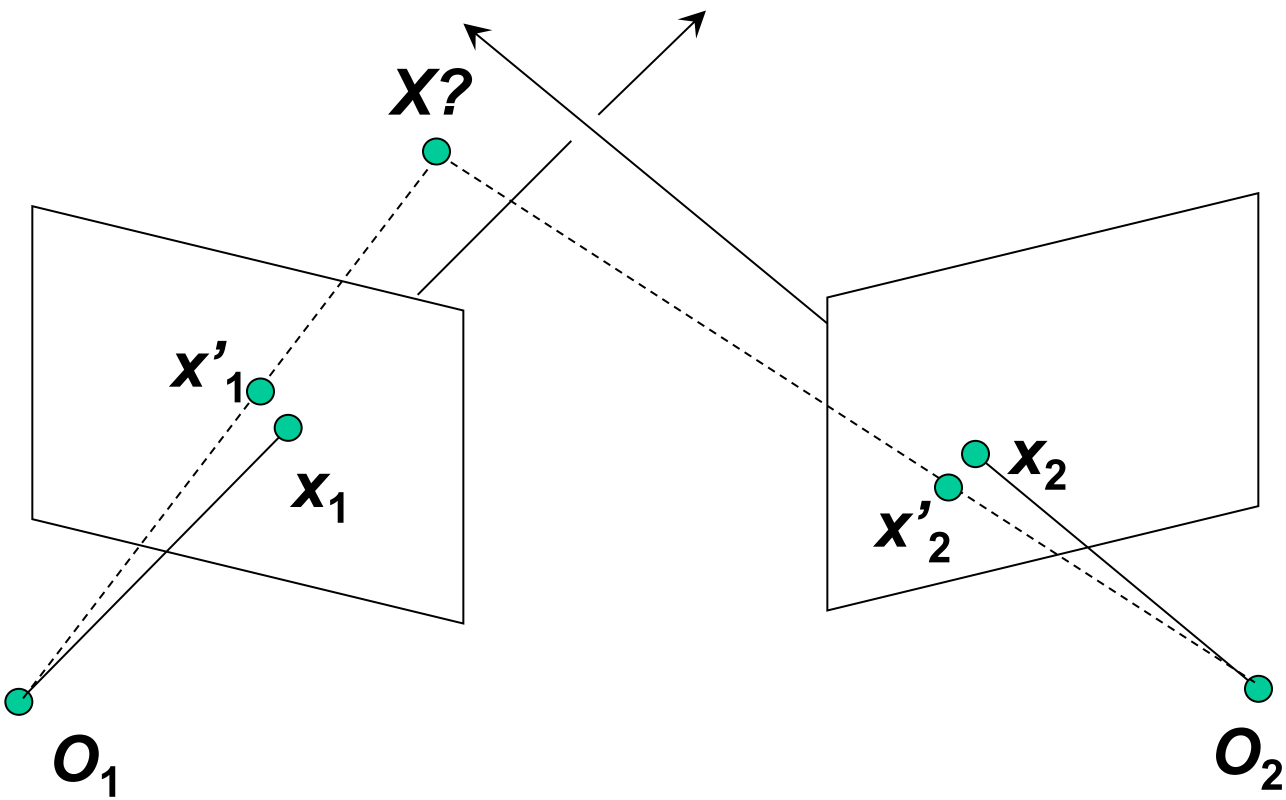
Triangulation: 3) Nonlinear Approach

- Find X that minimizes

$$d^2(x_1, P_1 X) + d^2(x_2, P_2 X)$$



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Triangulation: 3) Nonlinear Approach

- Find X that minimizes

$$d^2(x_1, P_1X) + d^2(x_2, P_2X)$$

- This approach is the most accurate, but unlike the other two methods, it doesn't have a closed-form solution.

- Iterative algorithm
  - Initialize with linear estimate.
  - Optimize with Gauss-Newton or Levenberg-Marquardt (see H&Z Appendix 6).

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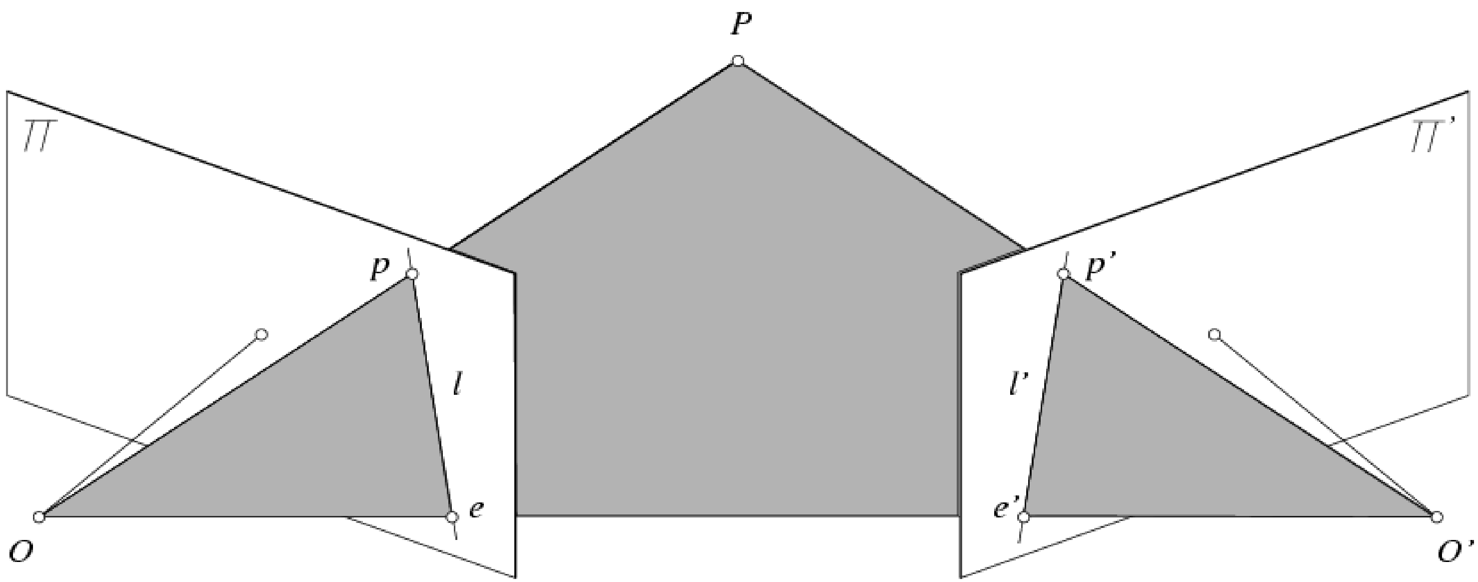
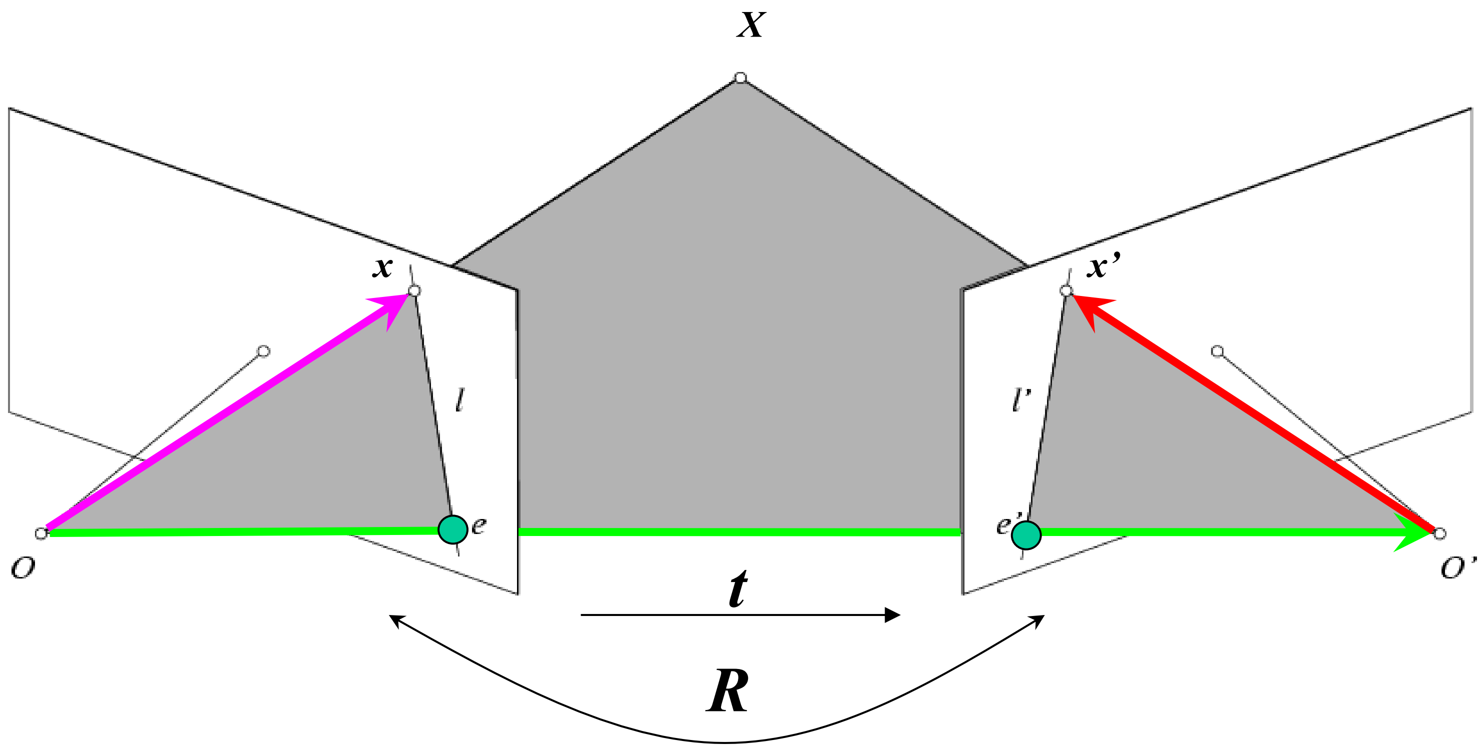
Revisiting Epipolar Geometry



- Let's look again at the epipolar constraint
  - For the calibrated case (but in homogenous coordinates)
  - For the uncalibrated case

Epipolar Geometry: Calibrated Case



Camera matrix: $[\mathbf{I}|\mathbf{0}]$

$X = (u, v, w, 1)^T$

$x = (u, v, w)^T$

Camera matrix: $[R^T \,|\, -R^T t]$

Vector x' in second coord. system has coordinates Rx' in the first one.

The vectors x, t, and Rx' are coplanar

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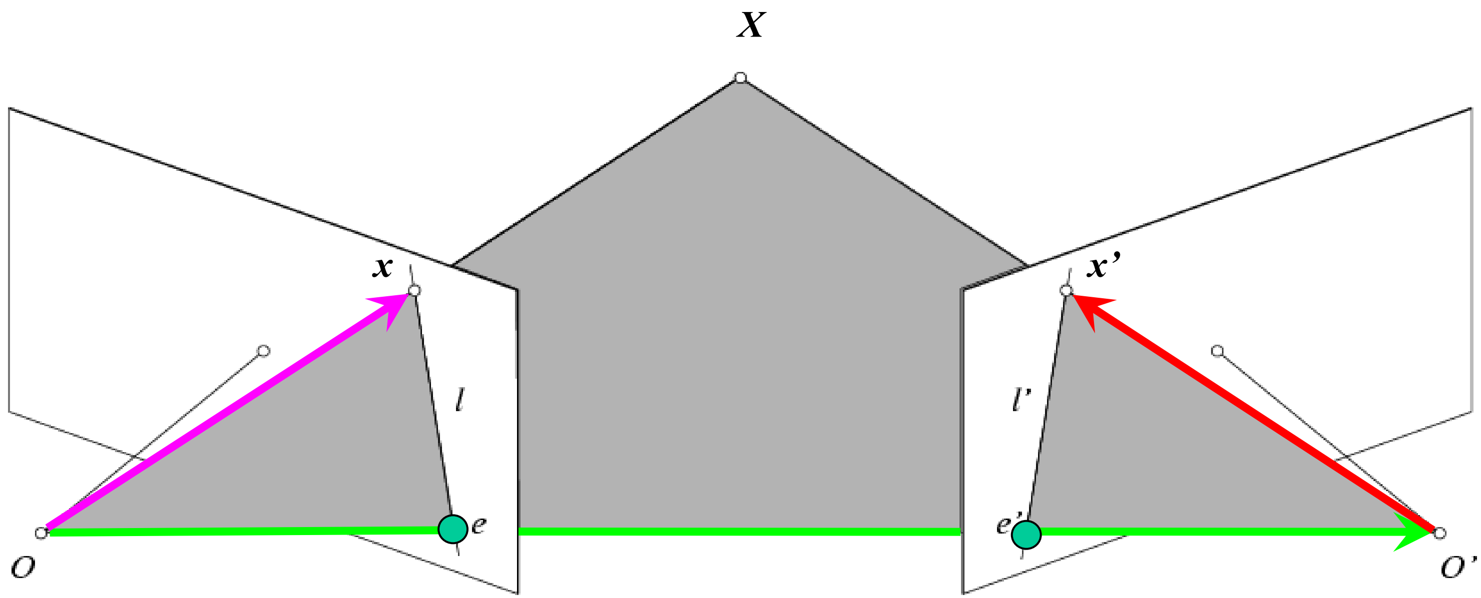
Epipolar Geometry: Calibrated Case



$$x \cdot [t \times (Rx')] = 0 \quad \Rightarrow \quad x^T E x' = 0 \quad \text{with} \quad E = [t_\times]R$$

Essential Matrix
(Longuet-Higgins, 1981)

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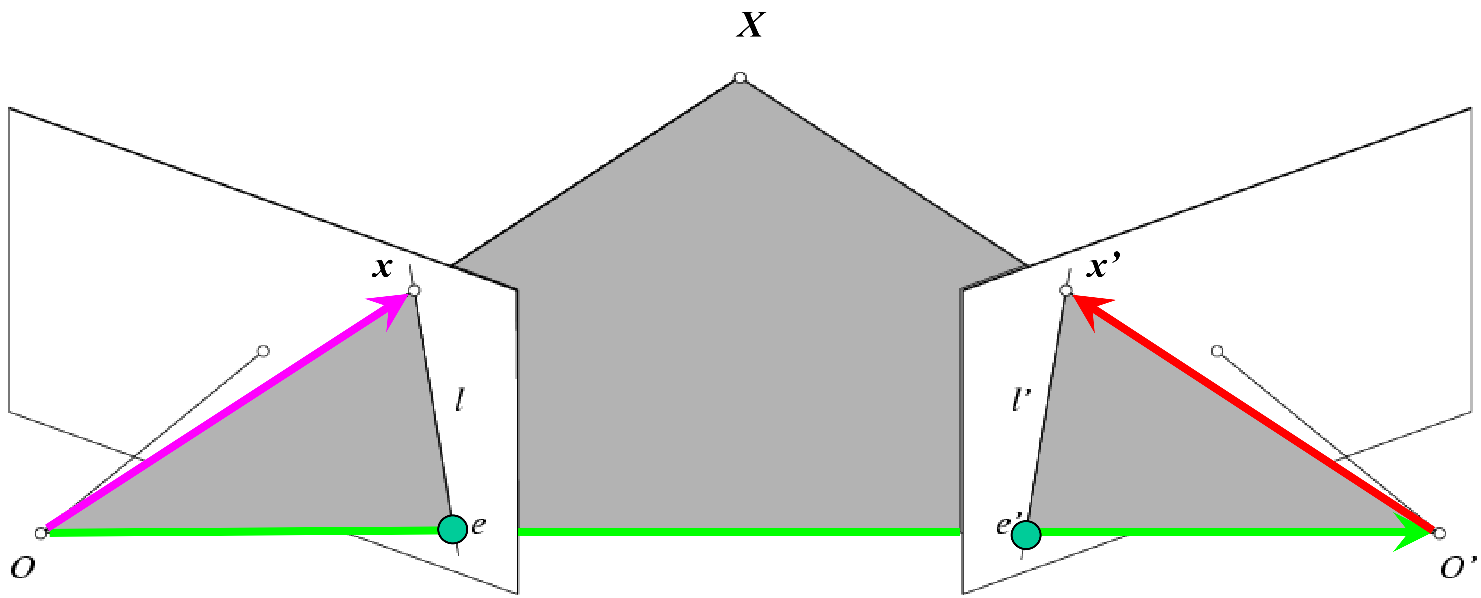
Epipolar Geometry: Calibrated Case



$$x \cdot [t \times (Rx')] = 0 \quad \Rightarrow \quad x^T E x' = 0 \quad \text{with} \quad E = [t_\times]R$$

- $E\,x'$ is the epipolar line associated with x' ($l = E\,x'$)
- $E^T x$ is the epipolar line associated with x ($l' = E^T x$)
- $E\,e' = 0$ and $E^T e = 0$ *Why?*
- E is singular (rank two) *Why?*
- E has five degrees of freedom (up to scale)

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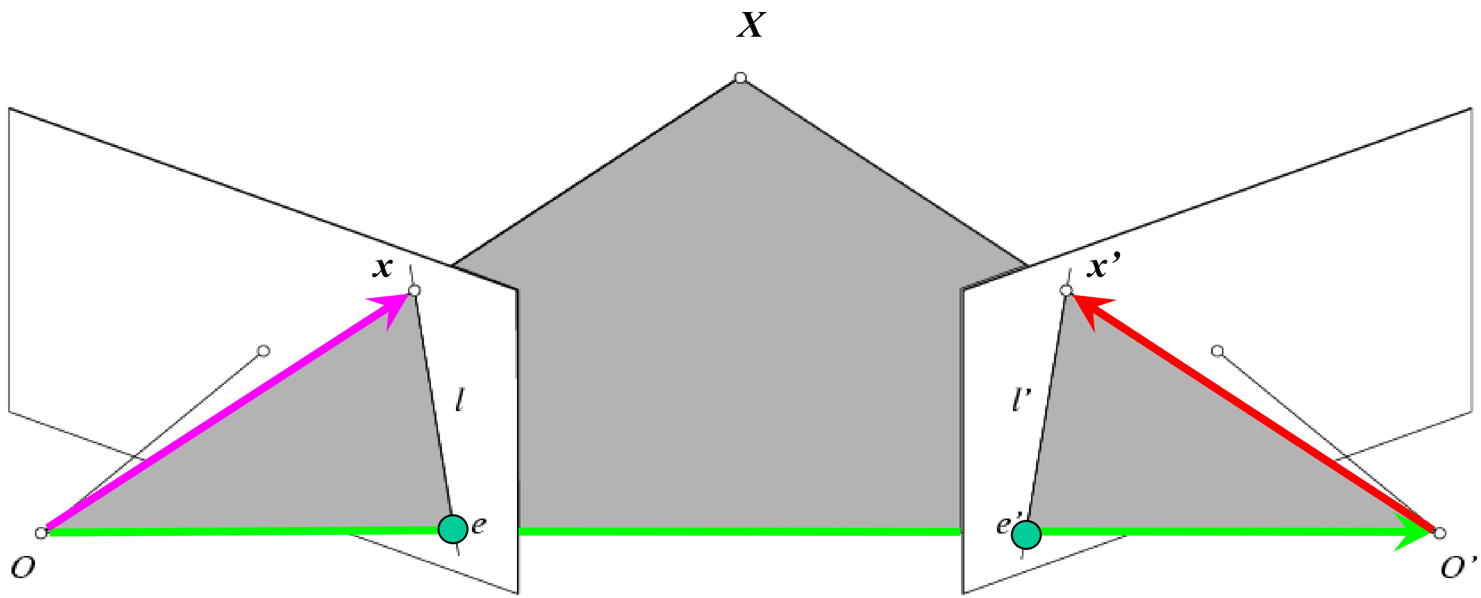
Epipolar Geometry: Uncalibrated Case



- The calibration matrices *K* and *K'* of the two cameras are unknown
- We can write the epipolar constraint in terms of *unknown* normalized coordinates:

$$\hat{x}^T E \hat{x}' = 0 \qquad\qquad x = K\hat{x}, \quad x' = K'\hat{x}'$$

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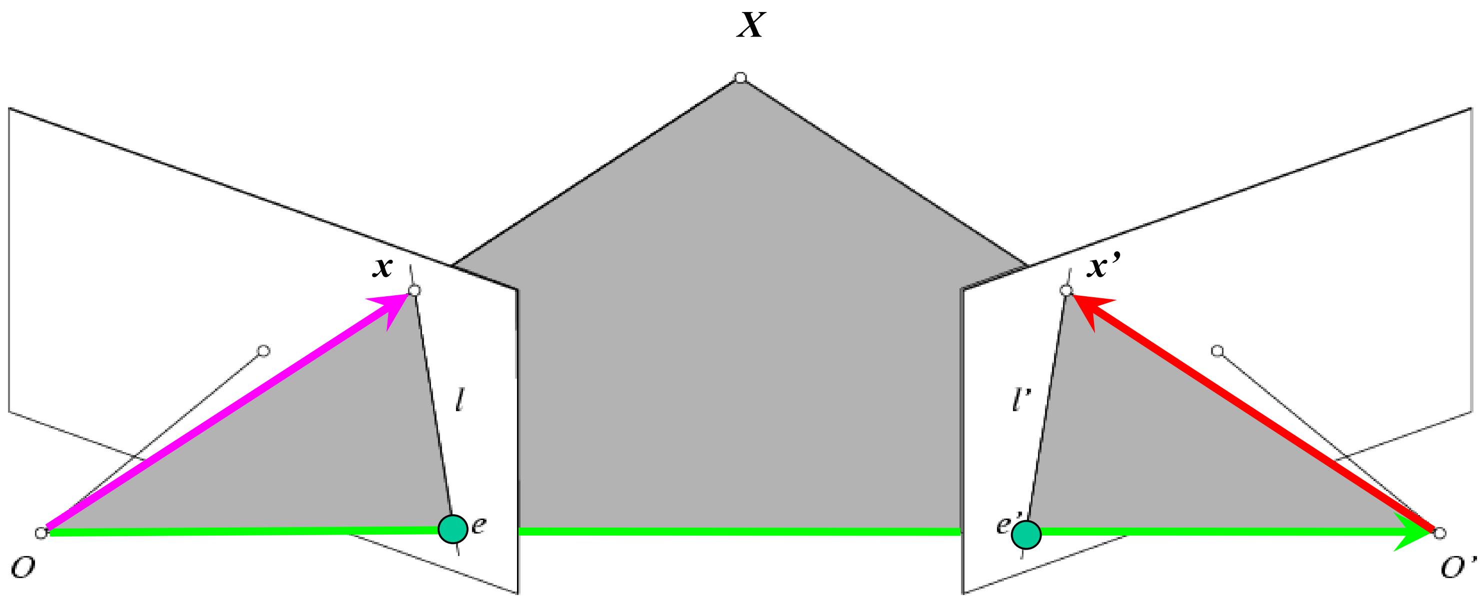
Epipolar Geometry: Uncalibrated Case



$$\hat{x}^T E \hat{x}' = 0 \quad \Rightarrow \quad x^T F x' = 0 \quad \text{with} \quad F = K^{-T} E K'^{-1}$$

$$x = K \hat{x}$$

$$x' = K' \hat{x}'$$

Fundamental Matrix
(Faugeras and Luong, 1992)

Computer Vision

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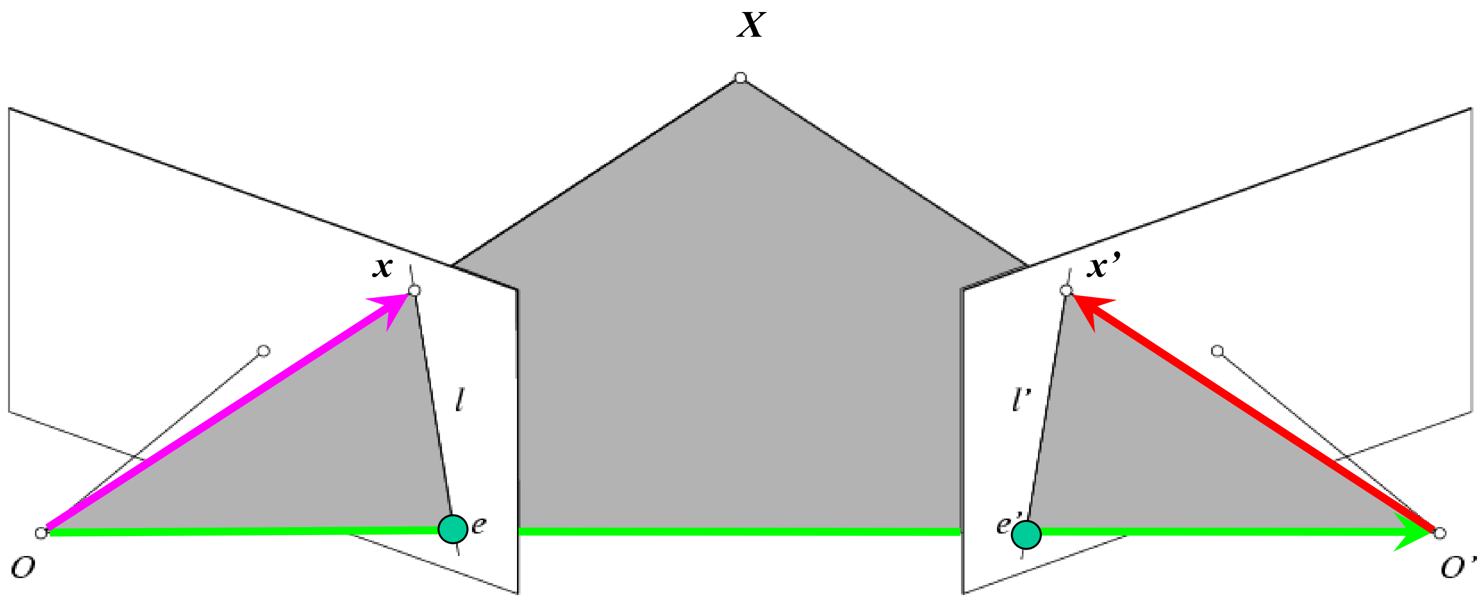
Epipolar Geometry: Uncalibrated Case



$$\hat{x}^T E \hat{x}' = 0 \;\Rightarrow\; x^T F x' = 0 \quad \text{with} \quad F = K^{-T} E K'^{-1}$$

- $F x'$ is the epipolar line associated with x' ($l = F x'$)
- $F^T x$ is the epipolar line associated with x ($l' = F^T x$)
- $F e' = 0$ and $F^T e = 0$
- F is singular (rank two)
- F has seven degrees of freedom

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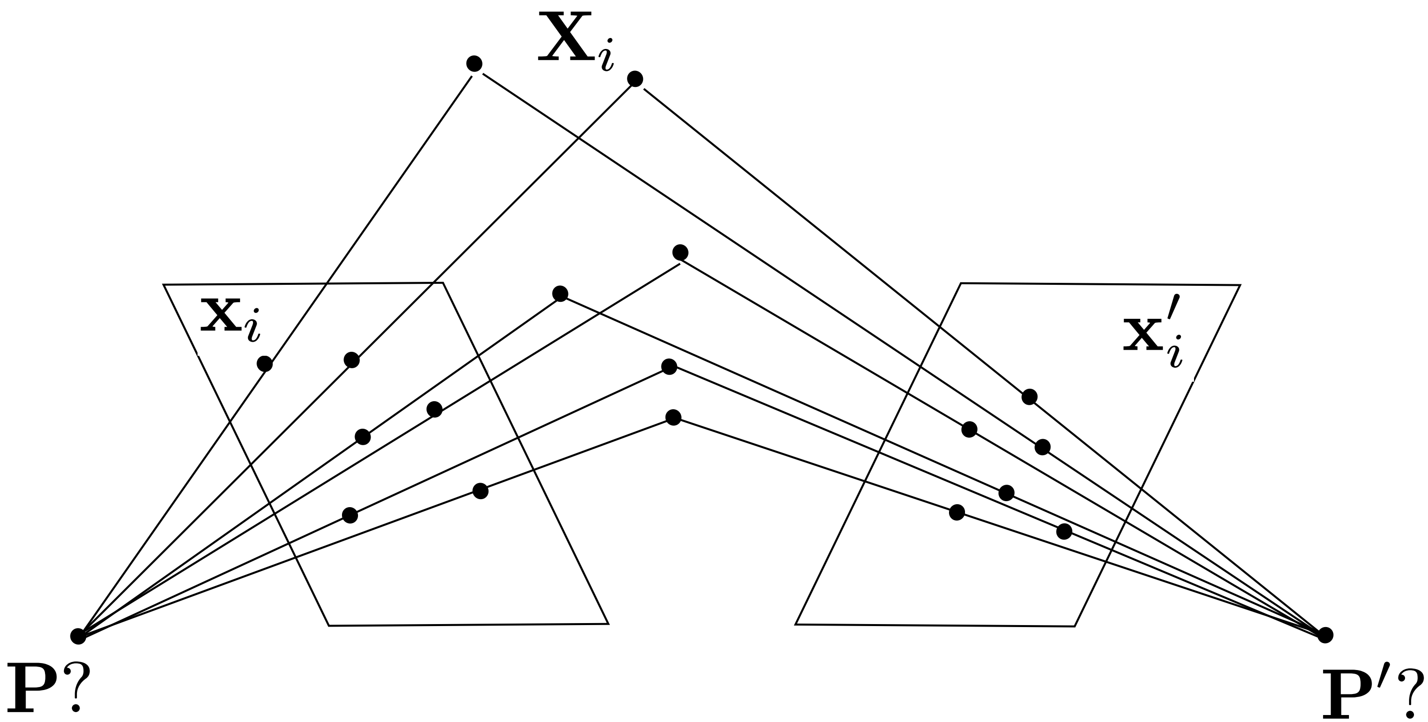
Estimating the Fundamental Matrix

- The Fundamental matrix defines the epipolar geometry between two uncalibrated cameras.
- How can we estimate F from an image pair?
  - We need correspondences…



The Eight-Point Algorithm

$\boldsymbol{x} = (\boldsymbol{u}, \boldsymbol{v}, \mathbf{1})^T, \quad \boldsymbol{x'} = (\boldsymbol{u'}, \boldsymbol{v'}, \mathbf{1})^T$

$$(u, v, 1)\begin{pmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{pmatrix}\begin{pmatrix} u' \\ v' \\ 1 \end{pmatrix} = 0 \quad \Rightarrow \quad [u'u, u'v, u', uv', vv', v', u, v, 1]\begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = 0$$

- Taking 8 correspondences:

$$\begin{bmatrix} u'_1u_1 & u'_1v_1 & u'_1 & u_1v'_1 & v_1v'_1 & v'_1 & u_1 & v_1 & 1 \\ u'_2u_2 & u'_2v_2 & u'_2 & u_2v'_2 & v_2v'_2 & v'_2 & u_2 & v_2 & 1 \\ u'_3u_3 & u'_3v_3 & u'_3 & u_3v'_3 & v_3v'_3 & v'_3 & u_3 & v_3 & 1 \\ u'_4u_4 & u'_4v_4 & u'_4 & u_4v'_4 & v_4v'_4 & v'_4 & u_4 & v_4 & 1 \\ u'_5u_5 & u'_5v_5 & u'_5 & u_5v'_5 & v_5v'_5 & v'_5 & u_5 & v_5 & 1 \\ u'_6u_6 & u'_6v_6 & u'_6 & u_6v'_6 & v_6v'_6 & v'_6 & u_6 & v_6 & 1 \\ u'_7u_7 & u'_7v_7 & u'_7 & u_7v'_7 & v_7v'_7 & v'_7 & u_7 & v_7 & 1 \\ u'_8u_8 & u'_8v_8 & u'_8 & u_8v'_8 & v_8v'_8 & v'_8 & u_8 & v_8 & 1 \end{bmatrix}\begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\mathbf{Af} = \mathbf{0}$$

Solve using… SVD!

This minimizes:

$$\sum_{i=1}^{N} (x_i^T F x_i')^2$$

Slide adapted from Svetlana Lazebnik

Excursion: Properties of SVD

- Frobenius norm
  - Generalization of the Euclidean norm to matrices

$$\|A\|_F = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n}|a_{ij}|^2} = \sqrt{\sum_{i=1}^{\min(m,n)}\sigma_i^2}$$

- Partial reconstruction property of SVD
  - Let σ_i $i=1,\ldots,N$ be the singular values of A.
  - Let $A_p = U_p D_p V_p^T$ be the reconstruction of A when we set $\sigma_{p+1}, \ldots, \sigma_N$ to zero.
  - Then $A_p = U_p D_p V_p^T$ is the best rank-p approximation of A in the sense of the Frobenius norm (i.e. the best least-squares approximation).

The Eight-Point Algorithm

- Problem with noisy data
  - The solution will usually not fulfill the constraint that F only has rank 2.

  $\Rightarrow$ *There will be no epipoles through which all epipolar lines pass!*

- Enforce the rank-2 constraint using SVD

$$F \overset{\text{SVD}}{=} \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^T = \boldsymbol{U}\begin{bmatrix} d_{11} & & \\ & d_{22} & \\ & & d_{33} \end{bmatrix}\begin{bmatrix} v_{11} & \cdots & v_{13} \\ \vdots & \ddots & \vdots \\ v_{31} & \cdots & v_{33} \end{bmatrix}^T$$

Set d_{33} to zero and reconstruct F

- As we have just seen, this provides the best least-squares approximation to the rank-2 solution.

Problem with the Eight-Point Algorithm

- In practice, this often looks as follows:

$$\begin{bmatrix} u'_1u_1 & u'_1v_1 & u'_1 & u_1v'_1 & v_1v'_1 & v'_1 & u_1 & v_1 & 1 \\ u'_2u_2 & u'_2v_2 & u'_2 & u_2v'_2 & v_2v'_2 & v'_2 & u_2 & v_2 & 1 \\ u'_3u_3 & u'_3v_3 & u'_3 & u_3v'_3 & v_3v'_3 & v'_3 & u_3 & v_3 & 1 \\ u'_4u_4 & u'_4v_4 & u'_4 & u_4v'_4 & v_4v'_4 & v'_4 & u_4 & v_4 & 1 \\ u'_5u_5 & u'_5v_5 & u'_5 & u_5v'_5 & v_5v'_5 & v'_5 & u_5 & v_5 & 1 \\ u'_6u_6 & u'_6v_6 & u'_6 & u_6v'_6 & v_6v'_6 & v'_6 & u_6 & v_6 & 1 \\ u'_7u_7 & u'_7v_7 & u'_7 & u_7v'_7 & v_7v'_7 & v'_7 & u_7 & v_7 & 1 \\ u'_8u_8 & u'_8v_8 & u'_8 & u_8v'_8 & v_8v'_8 & v'_8 & u_8 & v_8 & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Slide adapted from Svetlana Lazebnik

Problem with the Eight-Point Algorithm

- In practice, this often looks as follows:

250906.36	183269.57	921.81	200931.10	146766.13	738.21	272.19	198.81	1
2692.28	131633.03	176.27	6196.73	302975.59	405.71	15.27	746.79	1
416374.23	871684.30	935.47	408110.89	854384.92	916.90	445.10	931.81	1
191183.60	171759.40	410.27	416435.62	374125.90	893.65	465.99	418.65	1
48988.86	30401.76	57.89	298604.57	185309.58	352.87	846.22	525.15	1
164786.04	546559.67	813.17	1998.37	6628.15	9.86	202.65	672.14	1
116407.01	2727.75	138.89	169941.27	3982.21	202.77	838.12	19.64	1
135384.58	75411.13	198.72	411350.03	229127.78	603.79	681.28	379.48	1

$$\begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$\Rightarrow$ Poor numerical conditioning

$\Rightarrow$ Can be fixed by rescaling the data

Slide adapted from Svetlana Lazebnik

The Normalized Eight-Point Algorithm

1. Center the image data at the origin, and scale it so the mean squared distance between the origin and the data points is 2 pixels.
2. Use the eight-point algorithm to compute *F* from the normalized points.
3. Enforce the rank-2 constraint using SVD.

$$F \overset{\text{SVD}}{=} \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^T = \boldsymbol{U}\begin{bmatrix} d_{11} & & \\ & d_{22} & \\ & & d_{33} \end{bmatrix}\begin{bmatrix} v_{11} & \cdots & v_{13} \\ \vdots & \ddots & \vdots \\ v_{31} & \cdots & v_{33} \end{bmatrix}^T$$

Set d_{33} to zero and reconstruct F

4. Transform fundamental matrix back to original units: if T and T' are the normalizing transformations in the two images, then the fundamental matrix in original coordinates is $T^T F T'$.

Slide credit: Svetlana Lazebnik

[Hartley, 1995]

The Eight-Point Algorithm

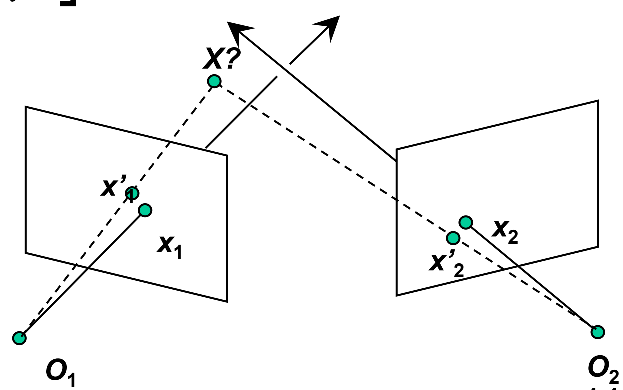
- Meaning of error $\sum_{i=1}^{N}(x_i^T F x_i')^2$:

  Sum of Euclidean distances between points x_i and epipolar lines Fx'_i (or points x'_i and epipolar lines $F^T x_i$), multiplied by a scale factor

- Nonlinear approach for refining the solution: minimize

$$\sum_{i=1}^{N}\left[\mathrm{d}^2(x_i, F x_i') + \mathrm{d}^2(x_i', F^T x_i)\right]$$

  - Similar to nonlinear minimization approach for triangulation.
  - Iterative approach (Gauss-Newton, Levenberg-Marquardt,…)



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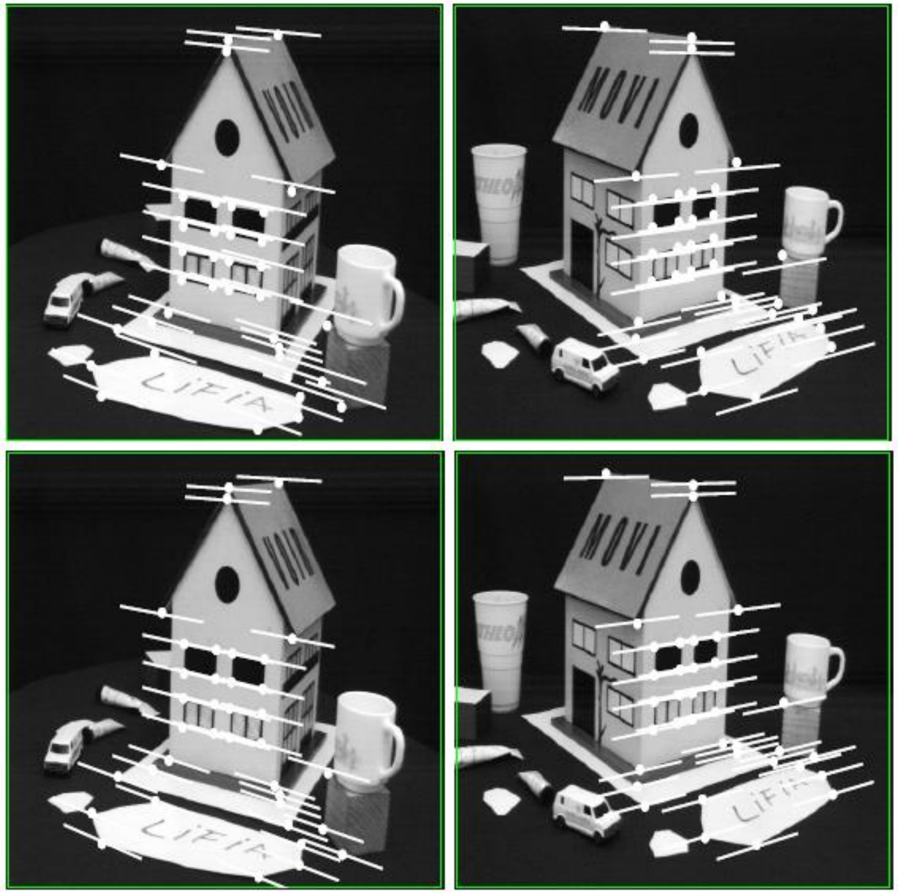
Comparison of Estimation Algorithms



	8-point	Normalized 8-point	Nonlinear least squares
Av. Dist. 1	2.33 pixels	0.92 pixel	0.86 pixel
Av. Dist. 2	2.18 pixels	0.85 pixel	0.80 pixel

Slide credit: Svetlana Lazebnik

3D Reconstruction with Weak Calibration

- Want to estimate world geometry without requiring calibrated cameras.
- Many applications:
  - Archival videos
  - Photos from multiple unrelated users
  - Dynamic camera system
- Main idea:
  - Estimate epipolar geometry from a (redundant) set of point correspondences between two uncalibrated cameras.

Slide credit: Kristen Grauman

Stereo Pipeline with Weak Calibration

- So, where to start with uncalibrated cameras?
  - Need to find fundamental matrix F *and* the correspondences (pairs of points (u',v') ↔ (u,v)).

- Procedure
  1. Find interest points in both images
  2. Compute correspondences
  3. Compute epipolar geometry
  4. Refine

Slide credit: Kristen Grauman

Example from Andrew Zisserman

Stereo Pipeline with Weak Calibration

1. Find interest points (e.g. Harris corners)

Slide credit: Kristen Grauman

Example from Andrew Zisserman

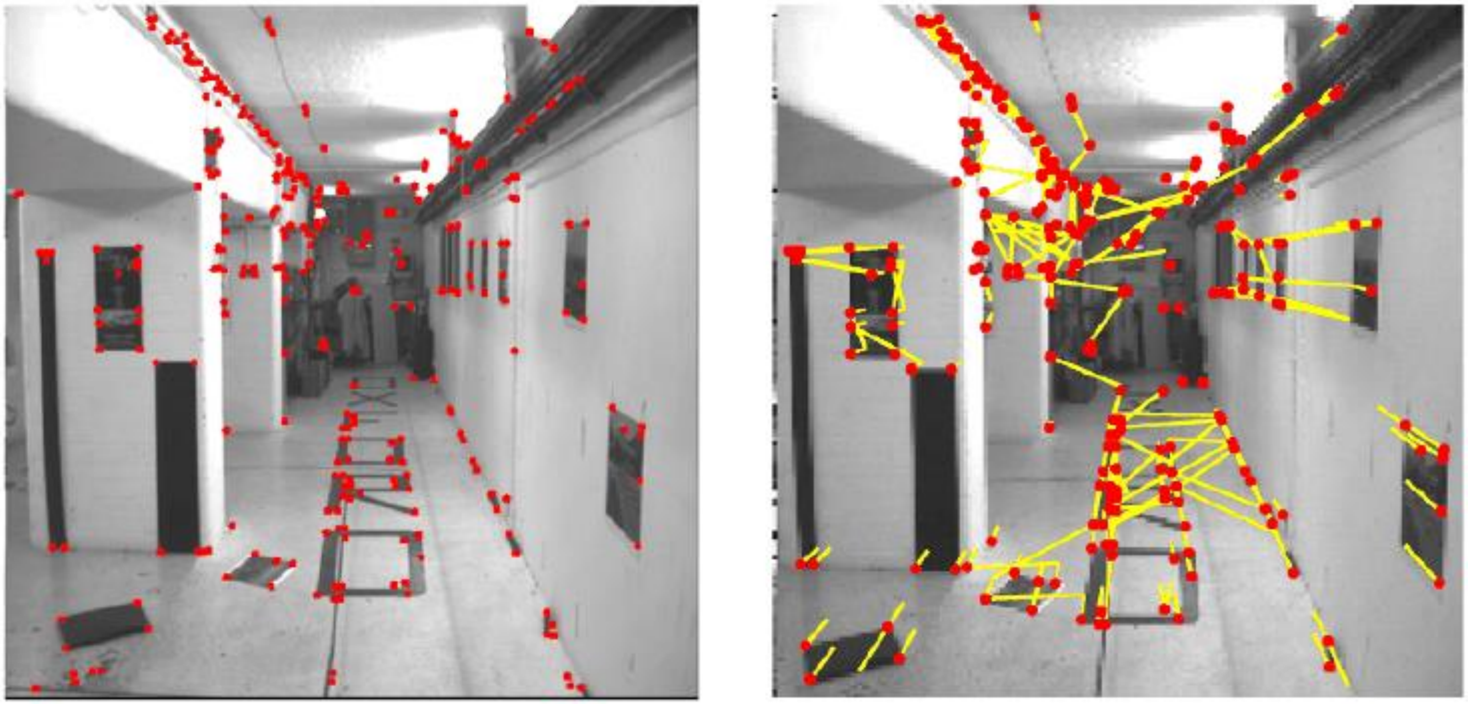
Stereo Pipeline with Weak Calibration

2. Match points using only proximity

Slide credit: Kristen Grauman

Example from Andrew Zisserman

Putative Matches based on Correlation Search



- Many wrong matches (10-50%), but enough to compute F

Example from Andrew Zisserman

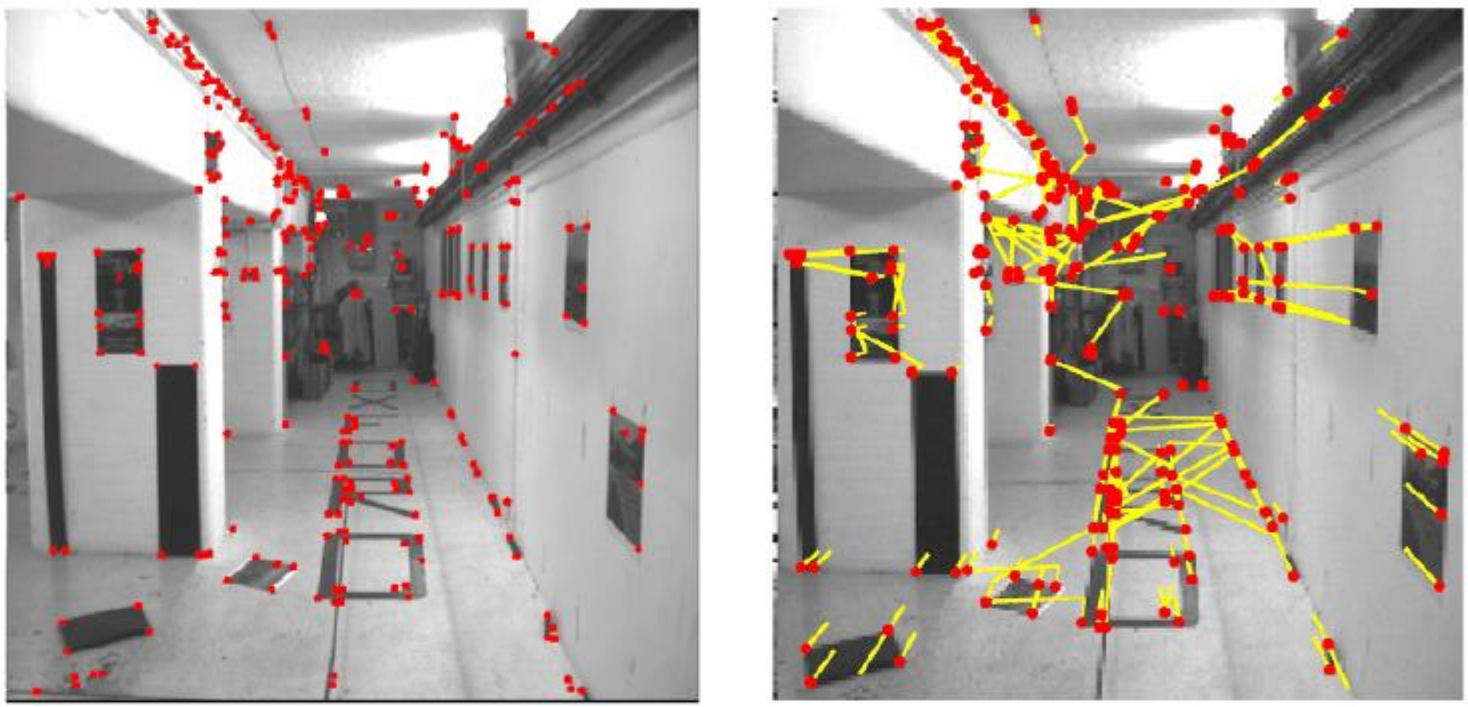
RANSAC for Robust Estimation of F

- Select random sample of correspondences
- Compute F using them
  - This determines epipolar constraint
- Evaluate amount of support – number of inliers within threshold distance of epipolar line
- Iterate until a solution with sufficient support has been found (or for max #iterations)

- Choose F with most support (#inliers)

Slide credit: Kristen Grauman

Putative Matches based on Correlation Search

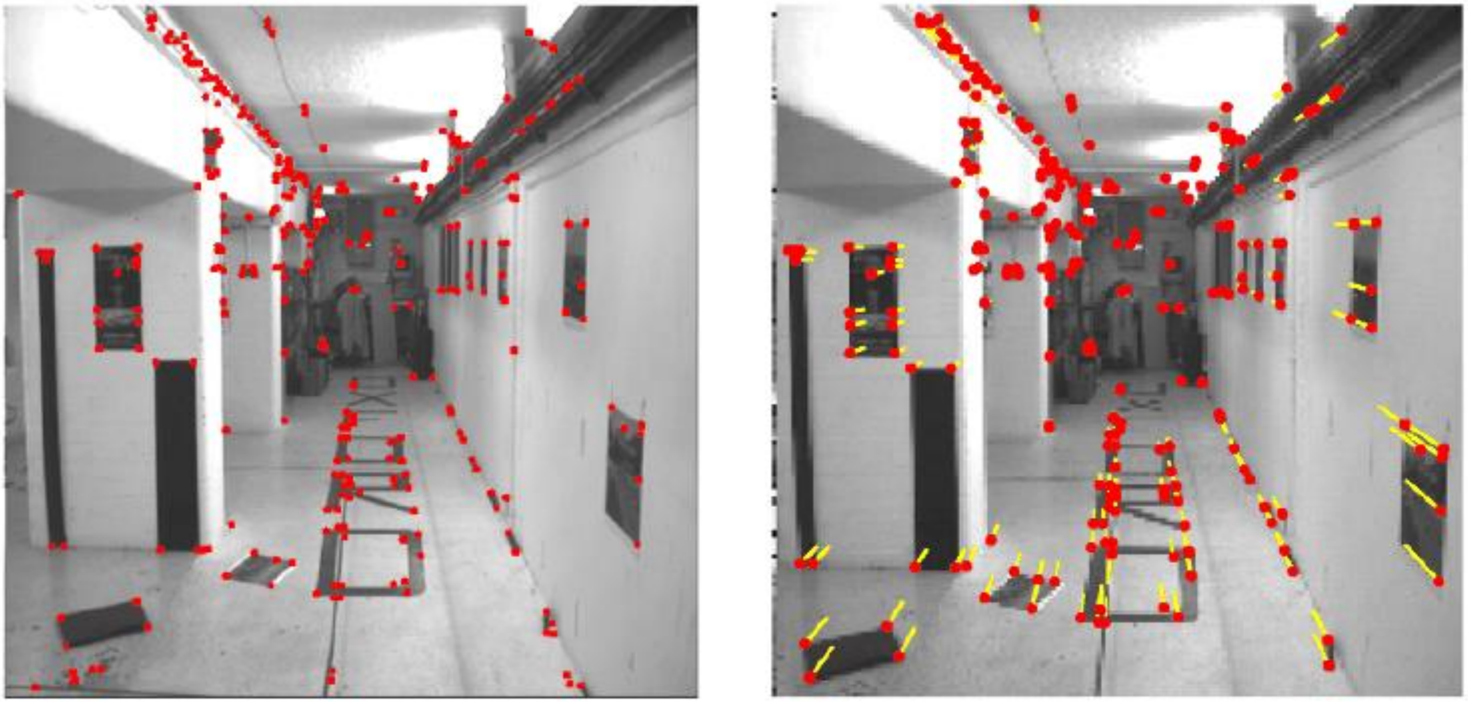


- Many wrong matches (10-50%), but enough to compute F

Example from Andrew Zisserman

Pruned Matches

- Correspondences consistent with epipolar geometry



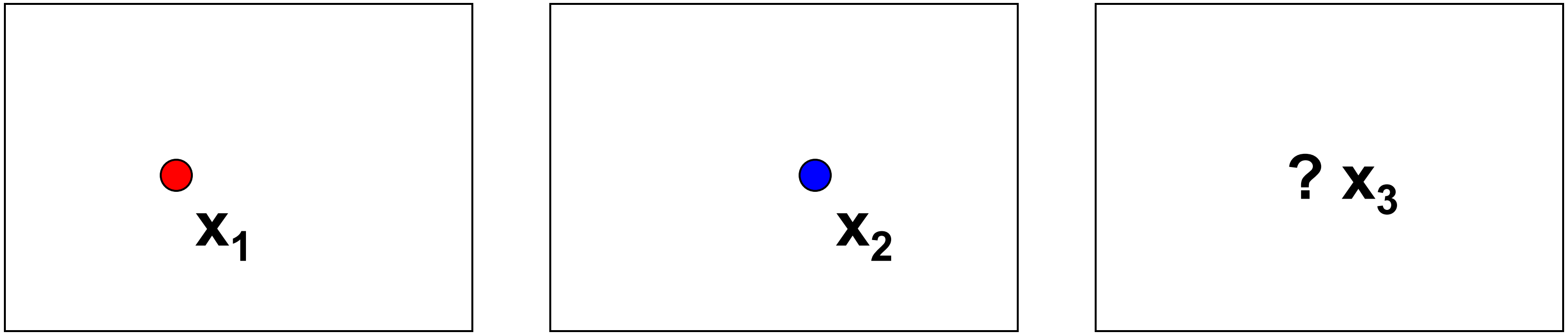
Example from Andrew Zisserman

Resulting Epipolar Geometry

Example from Andrew Zisserman

Extension: Epipolar Transfer

- Assume the epipolar geometry is known
- Given projections of the same point in two images, how can we compute the projection of that point in a third image?



Slide credit: Svetlana Lazebnik

Extension: Epipolar Transfer

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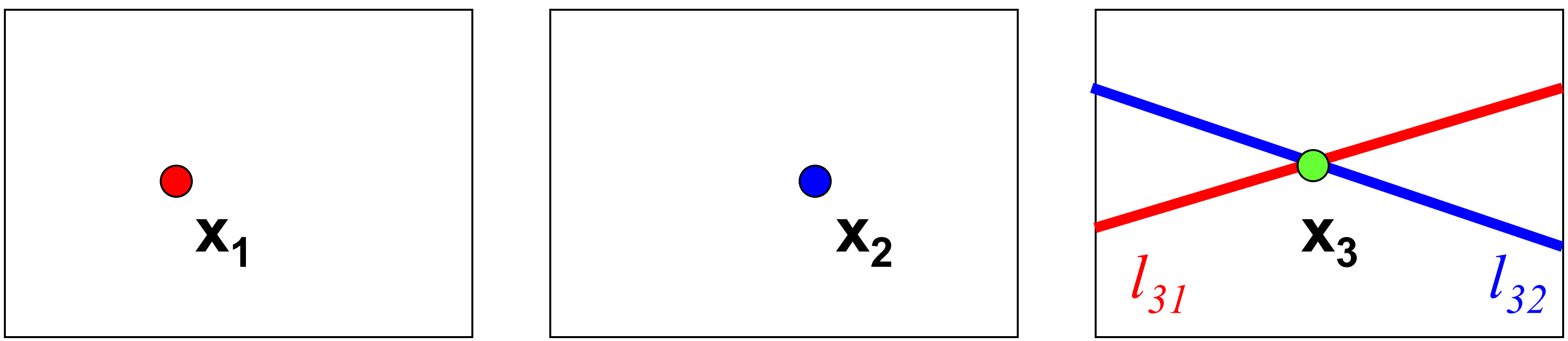


$$l_{31} = F^T{}_{13} x_1$$

$$l_{32} = F^T{}_{23} x_2$$

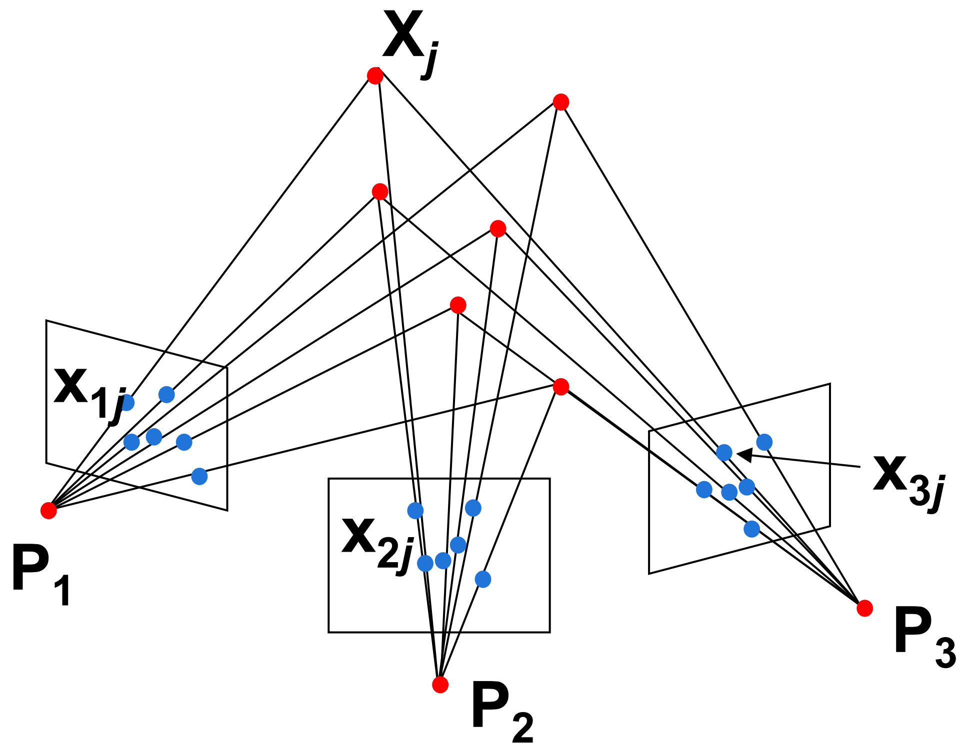
When does epipolar transfer fail?

Slide credit: Svetlana Lazebnik

Topics of This Lecture

- Camera calibration
  - Camera parameters
  - Calibration algorithm
- Revisiting Epipolar Geometry
  - Triangulation
  - Calibrated case: Essential matrix
  - Uncalibrated case: Fundamental matrix
  - Weak calibration
  - Epipolar Transfer
- Sneak Peek: Structure from Motion (SfM)
  - Motivation
  - Ambiguity

Structure from Motion



- Given: *m* images of *n* fixed 3D points

$$\mathrm{x}_{ij} = \mathrm{P}_i \mathrm{X}_j, \qquad i = 1, \dots, m, \quad j = 1, \dots, n$$

- Problem: estimate *m* projection matrices P_i and *n* 3D points X_j from the *mn* correspondences x_{ij}

Slide credit: Svetlana Lazebnik

Applications

- E.g., movie special effects

Video

Video Credit: Stefan Hafenegger

Structure from Motion Ambiguity

- If we scale the entire scene by some factor *k* and, at the same time, scale the camera matrices by the factor of 1/*k*, the projections of the scene points in the image remain exactly the same:

$$\mathbf{x} = \mathbf{P}\mathbf{X} = \left(\frac{1}{k}\mathbf{P}\right)(k\mathbf{X})$$

$\Rightarrow$ It is impossible to recover the absolute scale of the scene!

Slide credit: Svetlana Lazebnik

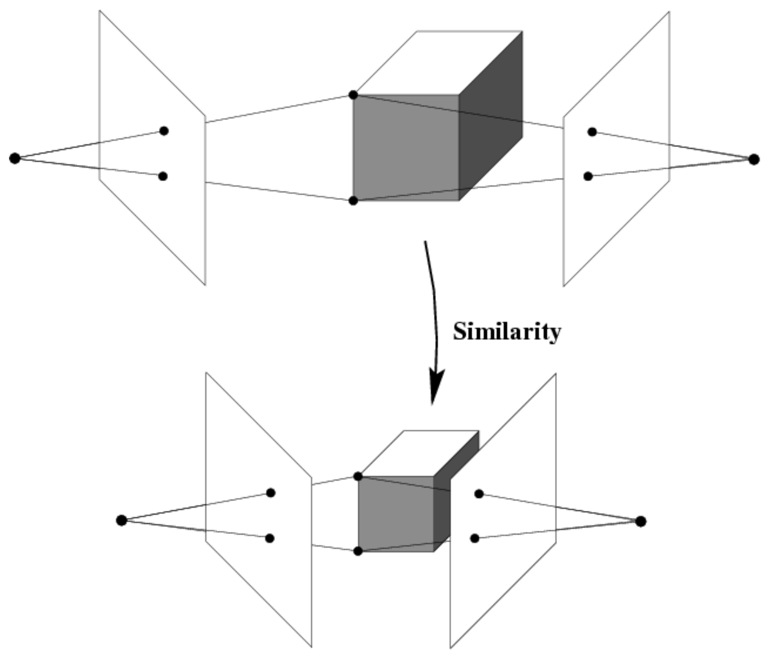
Structure from Motion Ambiguity

- If we scale the entire scene by some factor *k* and, at the same time, scale the camera matrices by the factor of 1/*k*, the projections of the scene points in the image remain exactly the same.
- More generally: if we transform the scene using a transformation Q and apply the inverse transformation to the camera matrices, then the images do not change

$$\mathbf{x} = \mathbf{P}\mathbf{X} = (\mathbf{P}\mathbf{Q}^{-1})\mathbf{Q}\mathbf{X}$$

Slide credit: Svetlana Lazebnik

Reconstruction Ambiguity: Similarity

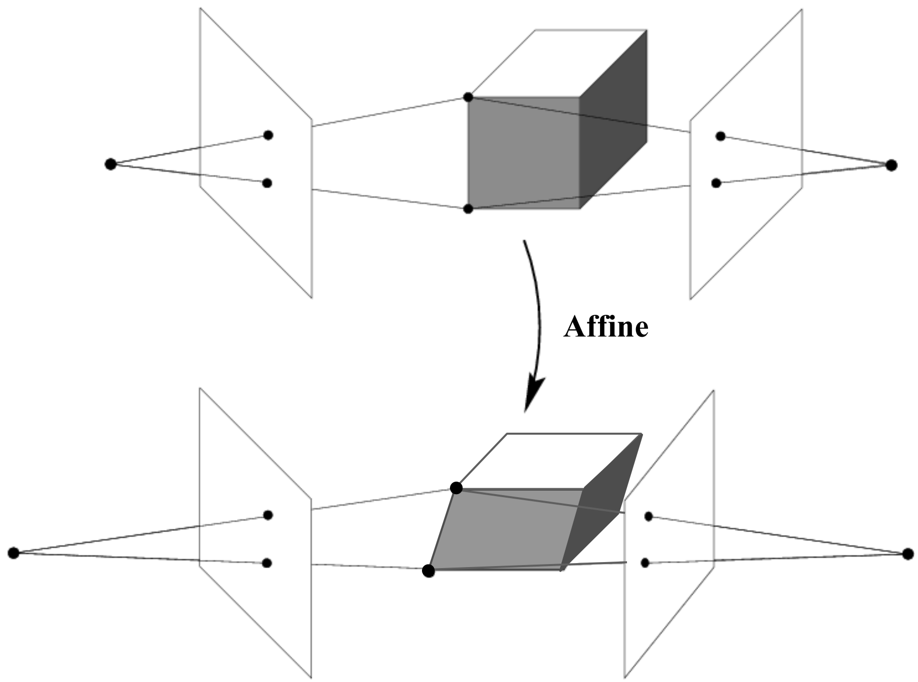


$$\mathbf{x} = \mathbf{P}\mathbf{X} = (\mathbf{P}\mathbf{Q}_S^{-1})\mathbf{Q}_S\mathbf{X}$$

Slide credit: Svetlana Lazebnik

Images from Hartley & Zisserman

Reconstruction Ambiguity: Affine

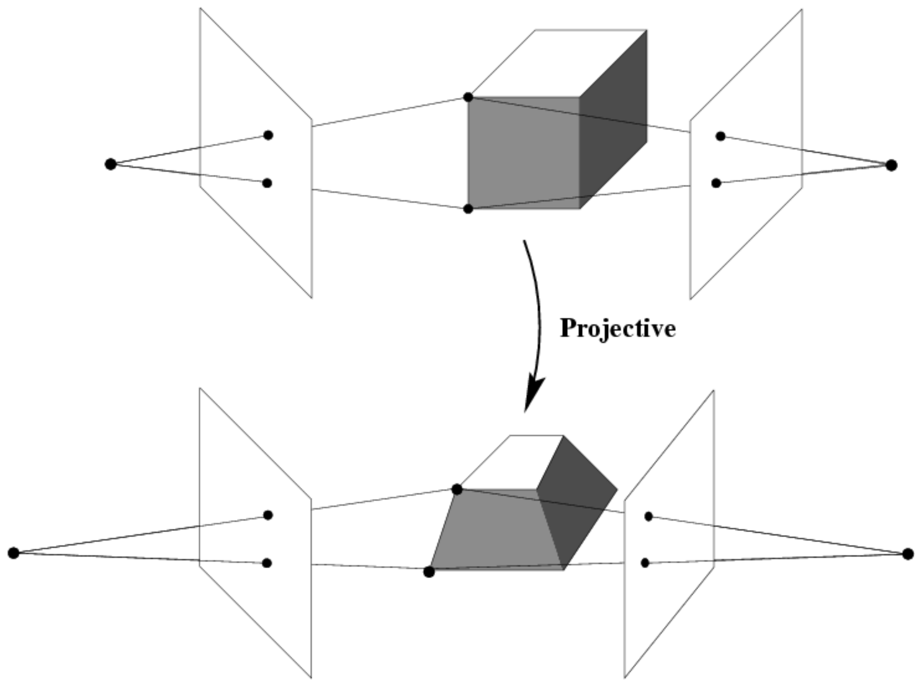


$$\mathbf{x} = \mathbf{P}\mathbf{X} = (\mathbf{P}\mathbf{Q}_A^{-1})\mathbf{Q}_A\mathbf{X}$$

Slide credit: Svetlana Lazebnik

Images from Hartley & Zisserman

Reconstruction Ambiguity: Projective

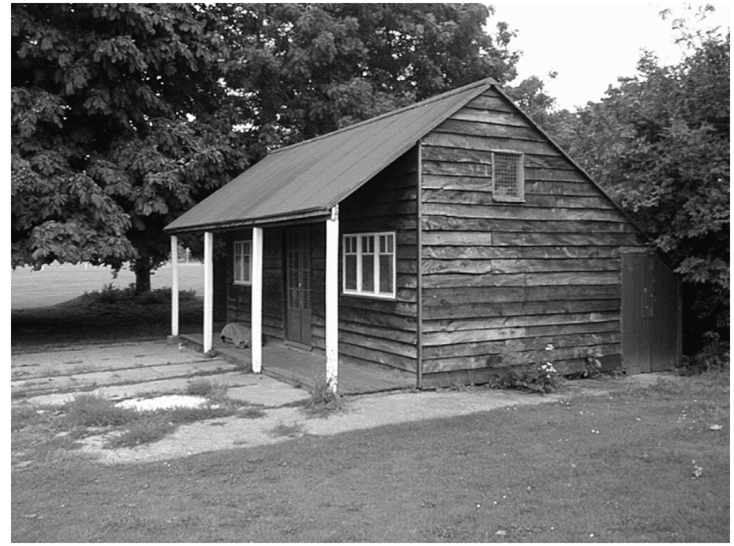


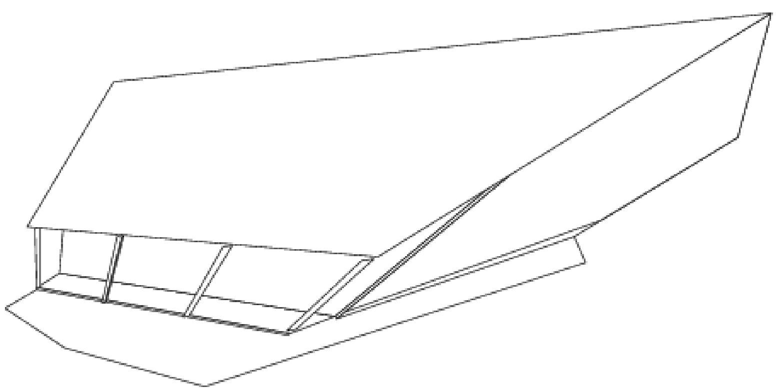
$$\mathbf{x} = \mathbf{P}\mathbf{X} = (\mathbf{P}\mathbf{Q}_P^{-1})\mathbf{Q}_P\mathbf{X}$$

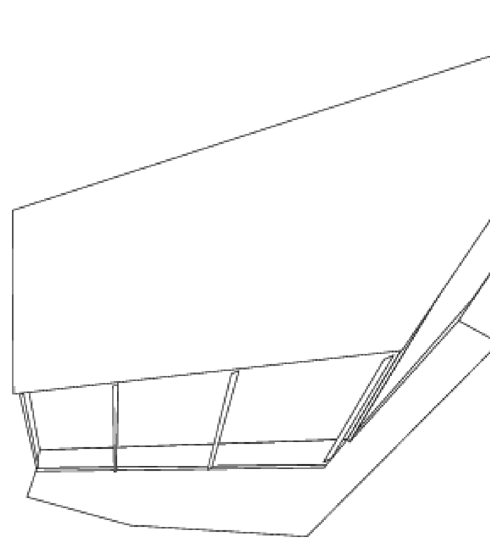
Slide credit: Svetlana Lazebnik

Images from Hartley & Zisserman

Projective Ambiguity



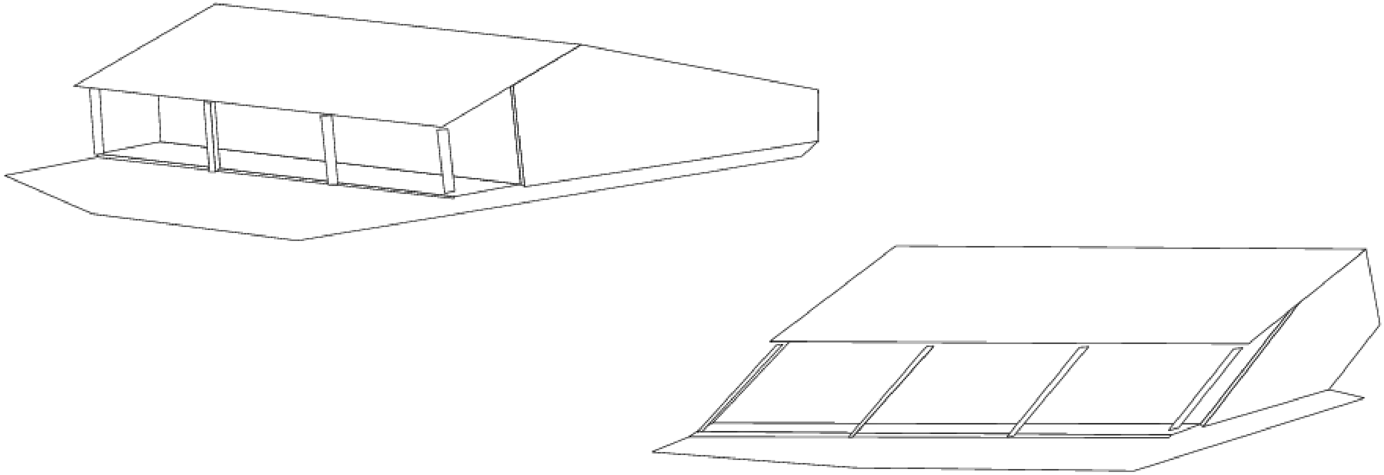




Slide credit: Svetlana Lazebnik

Images from Hartley & Zisserman

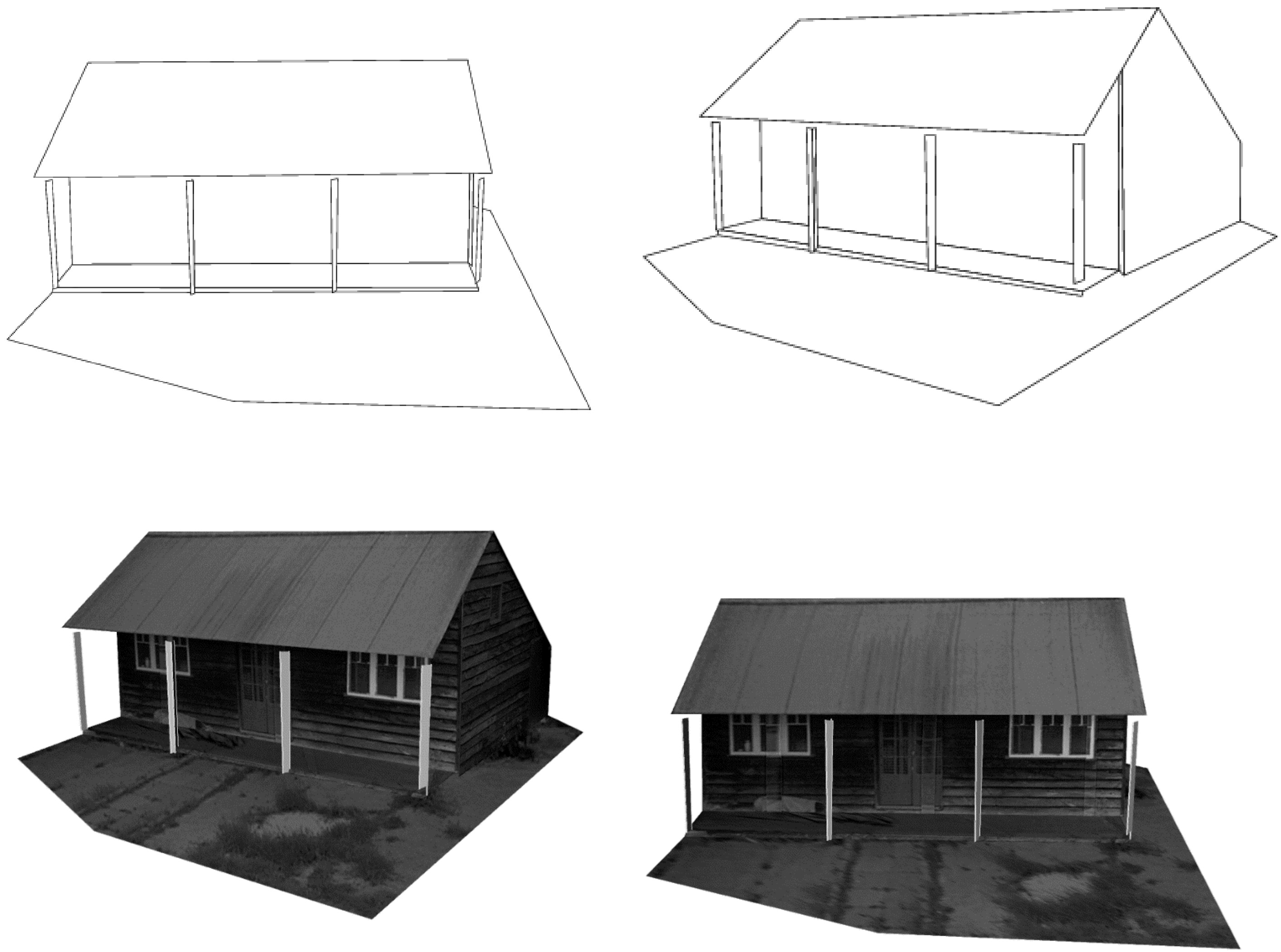
From Projective to Affine



Slide credit: Svetlana Lazebnik

Images from Hartley & Zisserman

From Affine to Similarity



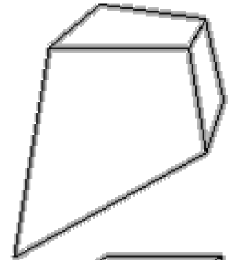
Slide credit: Svetlana Lazebnik

Images from Hartley & Zisserman

Hierarchy of 3D Transformations

Projective
15dof

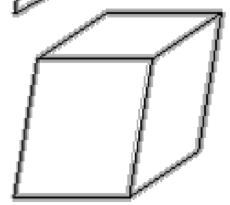
$$\begin{bmatrix} A & t \\ v^T & v \end{bmatrix}$$



Preserves intersection and tangency

Affine
12dof

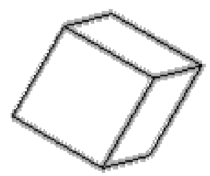
$$\begin{bmatrix} A & t \\ 0^T & 1 \end{bmatrix}$$



Preserves parallellism, volume ratios

Similarity
7dof

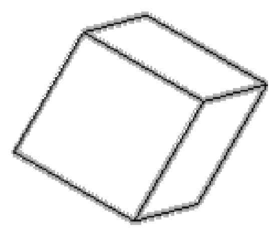
$$\begin{bmatrix} sR & t \\ 0^T & 1 \end{bmatrix}$$



Preserves angles, ratios of length

Euclidean
6dof

$$\begin{bmatrix} R & t \\ 0^T & 1 \end{bmatrix}$$



Preserves angles, lengths

- With no constraints on the camera calibration matrix or on the scene, we get a *projective* reconstruction.
- Need additional information to *upgrade* the reconstruction to affine, similarity, or Euclidean (e.g., square pixel assumption).

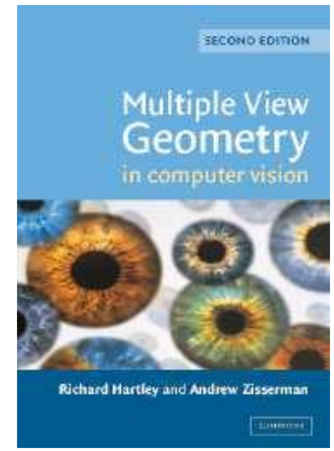
Slide credit: Svetlana Lazebnik

More about this in the Computer Vision 2 lecture…

References and Further Reading

- Background information on camera models and calibration algorithms can be found in Chapters 6 and 7 of

R. Hartley, A. Zisserman
Multiple View Geometry in Computer Vision
2nd Ed., Cambridge Univ. Press, 2004



- Also recommended: Chapter 9 of the same book on Epipolar geometry and the Fundamental Matrix and Chapter 11.1-11.6 on automatic computation of F.